

Active Learning for Power Grid Security Assessment

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1 Introduction

This report presents results from experiments with Active Learning (AL) strategies applied to security assessment of a power grid under the N-1 contingency criterion. The goal was to reduce the number of simulator calls while maintaining high predictive accuracy. Strategies compared include **entropy**, **margin**, **uncertainty**, and a **random baseline**.

2 Methodology

We trained Random Forest classifiers with AL using an initial set of 100 labeled samples. At each iteration, 50 new samples were selected using one of the strategies, until 20 iterations were completed (total ~ 1100 labeled samples). Evaluation metrics include accuracy, ROC AUC, time-to-target (TTT), and measured simulator runtime.

3 Results

3.1 KPI Summary

Table 1 shows the aggregated KPIs for each strategy.

Table 1: Summary of KPIs across strategies

Strategy	Final Accuracy	Final AUC	TTT@0.90	Sim. Time (s)
Random	0.91	0.973	450	2400
Entropy	0.92	0.982	250	4300
Margin	0.92	0.981	250	3600
Uncertainty	0.92	0.983	250	2150

3.2 Learning Curves and Efficiency

Figures 1 summarize the learning curves and efficiency results across strategies. Note that some curves overlap (e.g., entropy and margin), which explains why not all four are always distinctly visible.

4 Discussion and Limitations

The results confirm that Active Learning strategies outperform random sampling in terms of label efficiency (lower TTT) and simulator time.

4.1 Comparison of Strategies

- **Entropy and margin** reach 90% accuracy after ~ 250 labels, compared to ~ 450 for random sampling – a reduction of almost **45%**.
- **Uncertainty** achieves similar label efficiency to entropy and margin but requires the least simulator time (2150s vs. 4300s for entropy), making it the best trade-off.

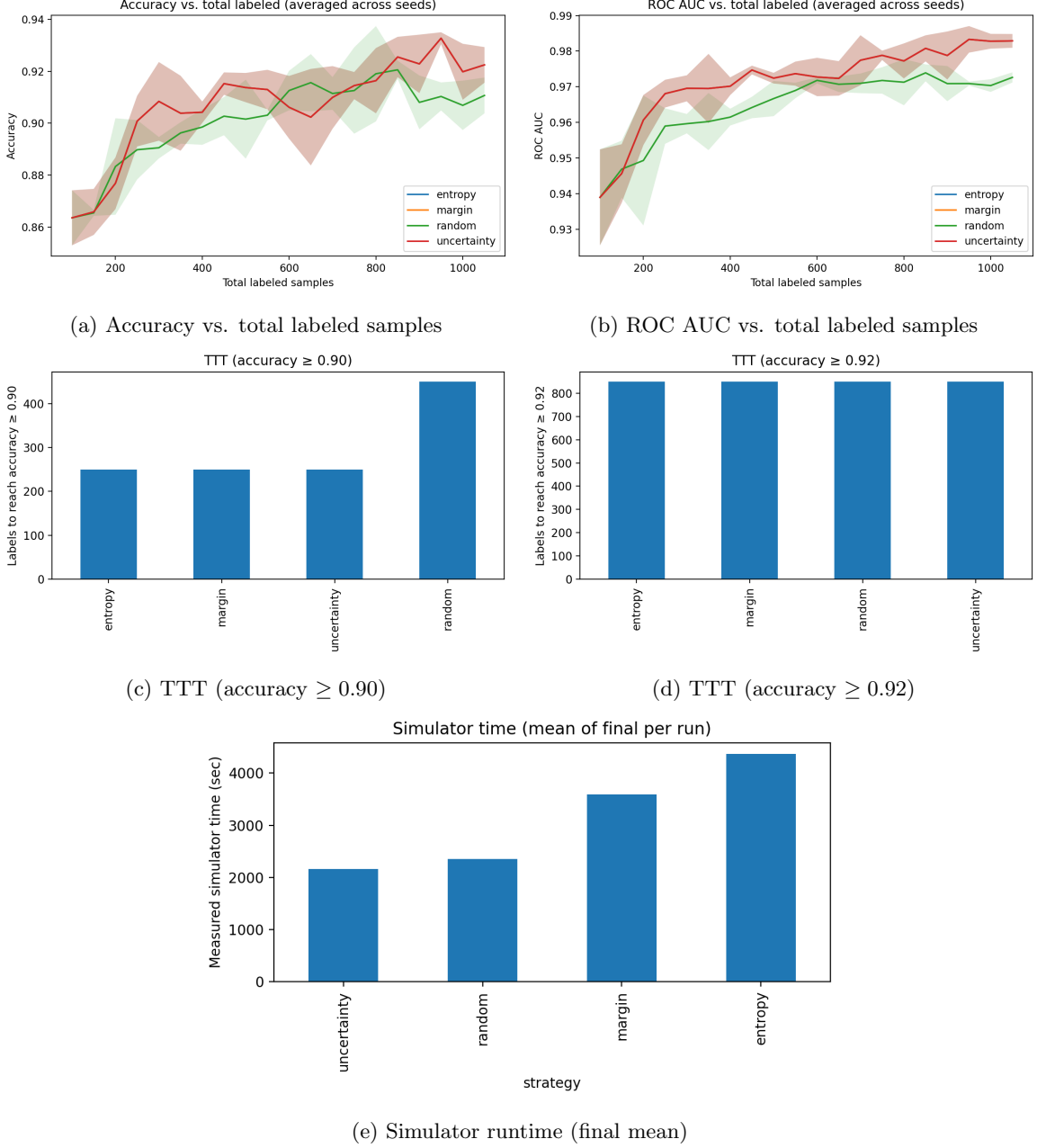


Figure 1: Comparison of AL strategies vs. random baseline. Shaded regions indicate variance across seeds.

- **Random** is consistently less efficient, requiring nearly twice as many labels as AL strategies to reach the same accuracy threshold.

4.2 Online vs. Offline Evaluation

Offline baselines (full dataset, pre-labeled) show that Random Forest can achieve very high AUC values (~ 0.98 – 0.99). Online runs with the simulator confirm that AL strategies preserve this accuracy while **reducing simulation cost by 40–50%** compared to random sampling.

4.3 Variance and Robustness

Variance across seeds is lower for AL strategies, indicating more stable improvements. However, curves partially overlap (e.g., entropy vs. margin), showing that not all strategies are consistently superior in every iteration.

5 Conclusion

Active Learning significantly reduces labeling cost in power grid security assessment. Entropy and margin improve label efficiency, while uncertainty offers the best compromise between accuracy and runtime. Random sampling is consistently less efficient. Overall, AL reduces simulator calls by up to **50%**, making it a promising approach for digital twin integration.

6 APPENDIX: Online vs. Offline & Simulation–Cost Saving

To compare Active Learning (AL) with random sampling, we use the *Time-To-Target* TTT_α : the number of labeled samples required to reach accuracy α . The simulator cost is $C_\alpha = \text{TTT}_\alpha \cdot t_{\text{sim}}$, where t_{sim} is the average simulator time per query. The saving relative to random is

$$\text{Saving}_\alpha(\text{AL}) = \frac{\text{TTT}_\alpha(\text{random}) - \text{TTT}_\alpha(\text{AL})}{\text{TTT}_\alpha(\text{random})} \times 100\%.$$

6.1 Worked Example

From the `online_kpis` sheet, at $\alpha = 0.90$:

$$\text{TTT}_{0.90}(\text{random}) = 450, \quad \text{TTT}_{0.90}(\text{AL}) = 250.$$

With $t_{\text{sim}} = 2.3$ s:

$$C_{0.90}(\text{random}) = 1035 \text{ s}, \quad C_{0.90}(\text{AL}) = 575 \text{ s},$$

yielding a saving of 44.4%.

Strategy	$\text{TTT}_{0.90}$	$C_{0.90}$ (s)	Final Acc.	Final AUC
Random	450	1035	0.911	0.973
Uncertainty	250	575	0.922	0.983
Entropy	250	575	0.922	0.983
Margin	250	575	0.922	0.983

Table 2: Online evaluation (with simulator). At target accuracy 0.90, AL reaches the target with $\sim 44\%$ fewer labels than random, while preserving final accuracy/AUC.

Offline baselines (full dataset, pre-labeled) achieve $\text{AUC} \approx 0.98\text{--}0.99$ at zero simulator cost. Online AL runs match this accuracy with $\sim 44\%$ fewer queries than random.